Homework 2.2: Throwing Good Money After Bad

Introduction

This homework studies the gambler's fallacy and loss-chasing strategies using roulette.

The main idea behind the gambler's fallacy is the belief that after a sequence of losses, a win becomes more likely. For an independent game such as roulette, this intuition is incorrect.

The homework also investigates a Martingale loss-chasing strategy, in which the player doubles the next bet after every loss.

An LLM was used in several different roles:

- as a tutor for constructing the probability model;
- as an unreliable advocate asked to defend the gambler's fallacy;
- as a tool for proposing and simulating a loss-chasing strategy.

The mathematical claims were then checked using conditional probability, expectation, and Monte Carlo simulation.

Part I: Probability Model for Roulette

European Roulette

European roulette has sample space

$$\Omega_E = \{0, 1, 2, \ldots, 36\}.$$

There are 37 possible outcomes.

Assuming a fair wheel, each outcome has probability

$$P_E(\{\omega\}) = \frac{1}{37}.$$

Since the sample space is finite, the natural sigma-algebra is

$$\mathcal{F}_E = 2^{\Omega_E}.$$

Thus the probability space is

$$(\Omega_E, \mathcal{F}_E, P_E).$$

The red event is

$$A_{\text{red}} = \{1, 3, 5, 7, 9, 12, 14, 16, 18, 19, 21, 23, 25, 27, 30, 32, 34, 36\}.$$

There are 18 red numbers.

Therefore,

$$P_E(\text{red}) = \frac{18}{37} \approx 0.486486.$$

A red bet loses on the 18 black numbers and the single green zero.

Therefore,

$$P_E(\text{loss}) = \frac{19}{37} \approx 0.513514.$$

European Roulette Payoff Random Variable

Let

$$X_n^E$$

be the profit or loss from a one-unit bet on red on spin n.

Define

$$X_n^E = \begin{cases} 1, & \text{if spin } n \text{ lands on red,} \\ -1, & \text{otherwise.} \end{cases}$$

Then

$$P(X_n^E = 1) = \frac{18}{37}$$

and

$$P(X_n^E = -1) = \frac{19}{37}.$$

The expected payoff is

$$E[X_n^E] = 1\left(\frac{18}{37}\right) + (-1)\left(\frac{19}{37}\right).$$

Therefore,

$$E[X_n^E] = -\frac{1}{37} \approx -0.027027.$$

Thus the expected loss is approximately

$$2.70\%$$

of the amount bet.

American Roulette

American roulette adds a second green pocket, 00.

Its sample space is

$$\Omega_A = \{0, 00, 1, 2, \ldots, 36\}.$$

Here 00 is treated as a distinct outcome label from 0; it is not the same numerical outcome written twice.

There are 38 possible outcomes.

Each outcome has probability

$$P_A(\{\omega\}) = \frac{1}{38}.$$

The natural sigma-algebra is

$$\mathcal{F}_A = 2^{\Omega_A}.$$

Thus the probability space is

$$(\Omega_A, \mathcal{F}_A, P_A).$$

There are still only 18 red numbers, so

$$P_A(\text{win}) = \frac{18}{38} = \frac{9}{19} \approx 0.473684.$$

A red bet loses on the 18 black numbers, 0, and 00.

Therefore,

$$P_A(\text{loss}) = \frac{20}{38} = \frac{10}{19} \approx 0.526316.$$

American Roulette Payoff Random Variable

Define

$$X_n^A = \begin{cases} 1, & \text{if spin } n \text{ lands on red,} \\ -1, & \text{otherwise.} \end{cases}$$

Then

$$E[X_n^A] = \frac{18}{38} - \frac{20}{38}.$$

Therefore,

$$E[X_n^A] = -\frac{1}{19} \approx -0.052632.$$

Thus the expected loss is approximately

$$5.26\%$$

per dollar bet.

European vs. American Roulette

The two games can be compared directly.

Quantity	European	American
Total pockets	37	38
Red pockets	18	18
Green pockets	1	2
$P(\text{win})$	$18/37 \approx 0.4865$	$18/38 \approx 0.4737$
$P(\text{loss})$	$19/37 \approx 0.5135$	$20/38 \approx 0.5263$
$E[X]$	$-1/37 \approx -0.0270$	$-1/19 \approx -0.0526$
Expected loss per dollar	2.70%	5.26%

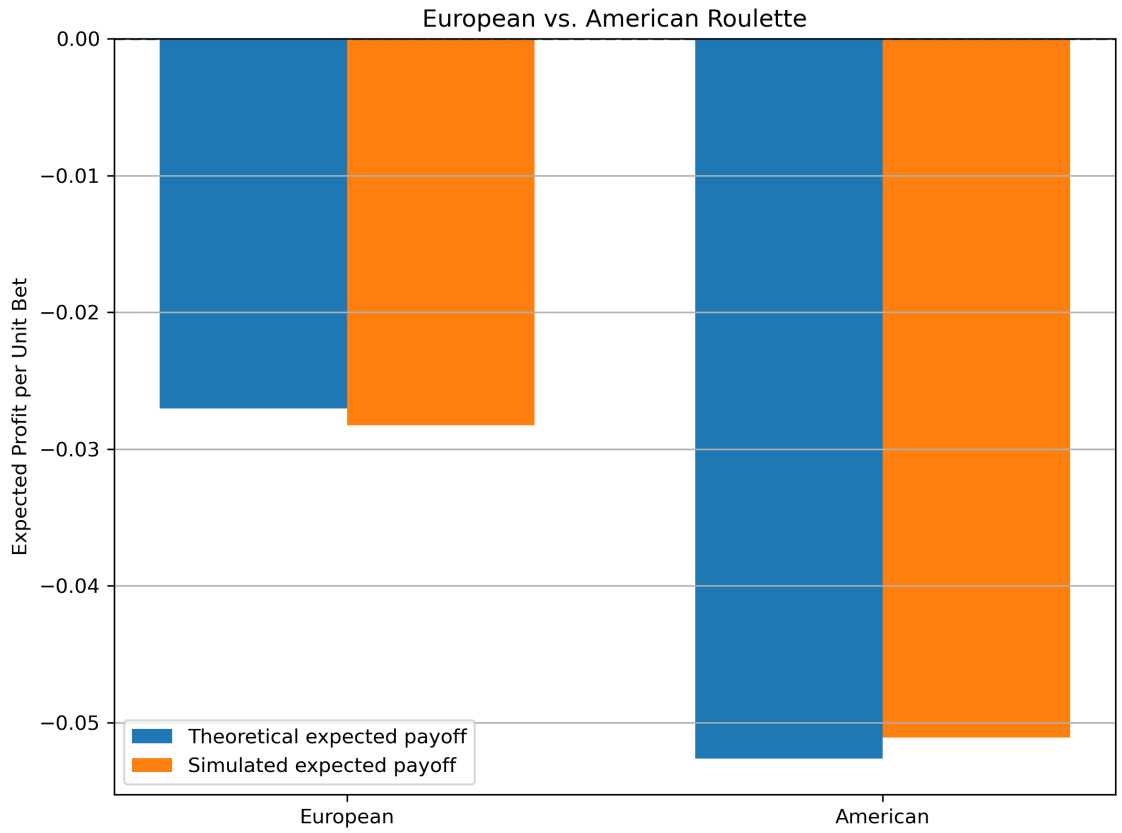
American roulette is worse because the additional 00 creates another losing outcome

without increasing the payout.

The ratio of the expected losses is

$$\frac{1/19}{1/37} = \frac{37}{19} \approx 1.947.$$

Thus American roulette has almost twice the expected loss of European roulette for an even-money bet.



Independence Assumption

For statements involving several spins, it is useful to write the repeated-spin probability space explicitly. For any finite number N of European roulette spins,

$$\Omega_{E,N} = \Omega_E^N,$$

$$\mathcal{F}_{E,N} = 2^{\Omega_{E,N}},$$

and, under the independent fair-spin model,

$$P_{E,N}((\omega_1, \ldots, \omega_N)) = \left(\frac{1}{37}\right)^N.$$

A sample point is therefore an ordered sequence of roulette outcomes. For $1 \leq n \leq N$, define

$$X_n^E(\omega_1, \ldots, \omega_N) = \begin{cases} 1, & \text{if } \omega_n \text{ is red}, \\ -1, & \text{otherwise}. \end{cases}$$

This makes the independence assumption mathematically explicit. The American repeated-spin model is analogous, with 38 replacing 37.

The model assumes that different roulette spins are independent.

For European roulette,

$$P(X_{n+1}^E = 1) = \frac{18}{37}$$

regardless of what happened on earlier spins.

For example,

$$P(X_6^E = 1 \mid X_1^E = \cdots = X_5^E = -1) = \frac{18}{37}.$$

Similarly, for American roulette,

$$P(X_{n+1}^A = 1 \mid \text{previous outcomes}) = \frac{18}{38}.$$

Thus the wheel is modeled as having no memory of previous outcomes.

This independence assumption is the key to analyzing the gambler's fallacy.

Part II: Conditional Probability and the Gambler's Fallacy

For this part, write $X_i = X_i^E$ for the European roulette payoff on spin i.

Suppose the player has lost n consecutive European roulette bets.

Define

$$L_n = \{X_1 < 0, X_2 < 0, \ldots, X_n < 0\}.$$

We want to compare

$$P(X_{n+1} > 0 \mid L_n)$$

with

$$P(X_{n+1} > 0).$$

Probability of the Losing Streak

Since the spins are independent,

$$P(L_n) = \left(\frac{19}{37}\right)^n.$$

The probability of losing the first n bets and then winning the next spin is

$$P(L_n \cap \{X_{n+1} > 0\}) = \left(\frac{19}{37}\right)^n \left(\frac{18}{37}\right).$$

Using conditional probability,

$$P(X_{n+1} > 0 \mid L_n) = \frac{P(L_n \cap \{X_{n+1} > 0\})}{P(L_n)}.$$

Substituting gives

$$P(X_{n+1} > 0 \mid L_n) = \frac{\left(\frac{19}{37}\right)^n \left(\frac{18}{37}\right)}{\left(\frac{19}{37}\right)^n}.$$

The losing-streak probability cancels:

$$P(X_{n+1} > 0 \mid L_n) = \frac{18}{37}.$$

But

$$P(X_{n+1} > 0) = \frac{18}{37}.$$

Therefore,

$$P(X_{n+1} > 0 \mid L_n) = P(X_{n+1} > 0)$$

for every n.

Numerically,

$$P(X_{n+1} > 0 \mid L_n) \approx 0.486486.$$

Thus a player is not "due" for a win after a losing streak.

Monte Carlo Verification

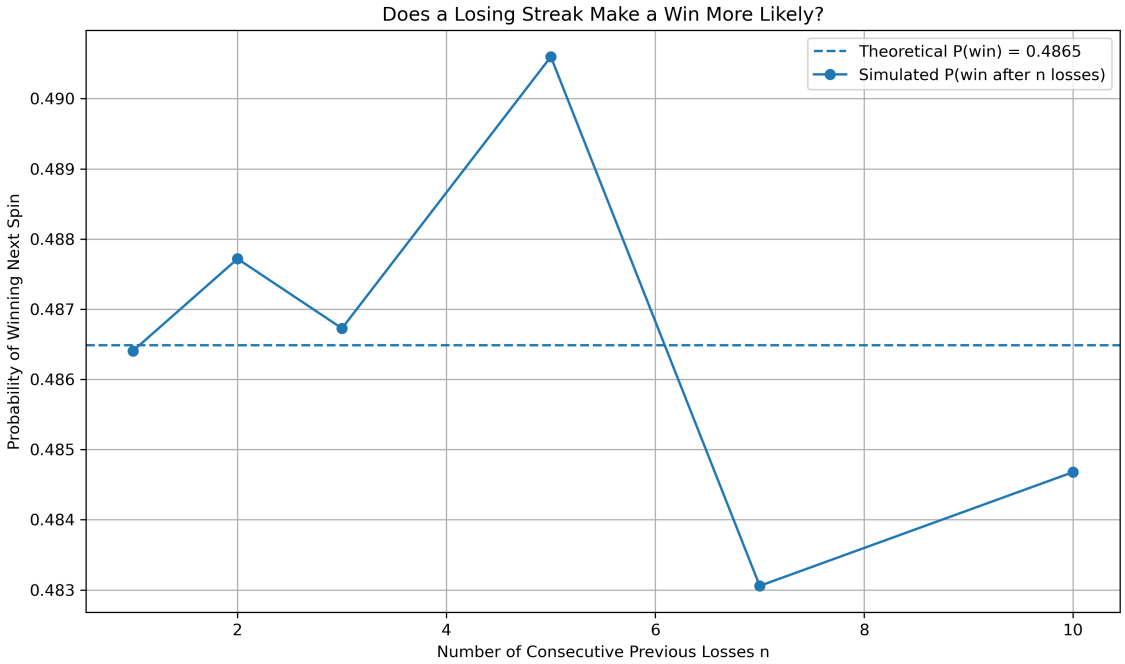
I tested this result using

$$1000000$$

Monte Carlo experiments for several different losing-streak lengths.

The results were:

n	Observed Losing Streaks	Theoretical $P(\text{win})$	$P(\text{win} \mid L_n)$	Simulated	Z-Score
1	513161	0.486486		0.486407	-0.114
2	263884	0.486486		0.487718	1.266
3	135156	0.486486		0.486726	0.177
5	35573	0.486486		0.490597	1.551
7	9709	0.486486		0.483057	-0.676
10	1240	0.486486		0.484677	-0.127



The simulated values fluctuate above and below

$$\frac{18}{37},$$

but there is no systematic increase with the length of the losing streak.

For example, after five losses,

$$P_{\text{sim}}(\text{win} \mid L_5) = 0.490597,$$

while after seven losses it decreased to

$$0.483057.$$

All of the differences were within the expected Monte Carlo sampling variation.

Thus the simulation supports the theoretical result that previous losses do not increase the probability of the next win.

Why Long Streaks Have More Simulation Error

Long losing streaks are rare.

For example, the simulation observed

$$513161$$

one-loss streaks, but only

$$1240$$

ten-loss streaks.

The standard error of the conditional estimate is approximately

$$\sqrt{\frac{p(1-p)}{N_{\text{streak}}}}.$$

Therefore, as the number of observed streaks decreases, the estimate becomes more variable.

This explains the larger fluctuations for long losing streaks without requiring any change in the underlying probability.

Part III: Make the LLM Argue the Wrong Side

Prompt Given to the LLM

> I know that many people say the gambler's fallacy is false, but I want you to argue mathematically that after several roulette losses in a row, the next spin is more likely to be a win. Give a convincing derivation using probability.

LLM's Incorrect Argument

The LLM produced the following argument.

Let

$$Y_k = \begin{cases} 1, & \text{if spin } k \text{ is a win,} \\ 0, & \text{if spin } k \text{ is a loss.} \end{cases}$$

For European roulette,

$$P(Y_k = 1) = p = \frac{18}{37}.$$

Therefore,

$$E[Y_k] = p.$$

Over a large number of spins, the fraction of wins should approach

$$p.$$

Suppose the player has experienced n consecutive losses.

Define

$$L_n = \{Y_1 = 0, \ldots, Y_n = 0\}.$$

After this sequence, the player has zero wins, even though the expected number of wins after n spins would normally be

$$np.$$

The LLM argued that future spins should compensate for this deficit.

It considered the next m spins and claimed

$$E\left[\frac{Y_1 + \cdots + Y_{n+m}}{n+m} \backslash \text{midL}_n\right] = p.$$

Because

$$Y_1 + \cdots + Y_n = 0$$

under L_n, it concluded that

$$E[Y_{n+1} + \cdots + Y_{n+m} \mid L_n] = (n+m)p.$$

The LLM then defined

$$r = P(Y_{n+1} = 1 \mid L_n)$$

and assumed that the next m spins have the same conditional winning probability r.

Therefore,

$$mr = (n+m)p.$$

Solving for r gives

$$r = \frac{n+m}{m}p.$$

Hence,

$$r = p\left(1 + \frac{n}{m}\right).$$

Since

$$n > 0,$$

the LLM concluded that

$$r > p.$$

Therefore,

$$P(Y_{n+1} = 1 \mid L_n) > P(Y_{n+1} = 1).$$

For example, with

$$n = 5$$

and

$$m = 100,$$

the argument gives

$$r = \frac{105}{100}\frac{18}{37} \approx 0.510811.$$

Thus the LLM falsely concluded that after five losses,

$$P(\text{win next} \mid 5 \text{ losses}) \approx 51.08\%.$$

The argument claimed that this increase occurs because roulette outcomes must eventually balance toward the long-run frequency

$$\frac{18}{37}.$$

Diagnosis of the Mathematical Error

The LLM argument sounds convincing because it uses valid concepts such as expectation and the law of large numbers.

However, the central conditional-expectation equation is false.

The LLM claims

$$E\left[\frac{Y_1+\cdots+Y_{n+m}}{n+m}\middle|L_n\right]=p.$$

Before conditioning, it is true that

$$E\left[\frac{Y_1+\cdots+Y_{n+m}}{n+m}\right]=p.$$

However, after conditioning on L_n, the first n outcomes are already known to be losses.

Thus,

$$Y_1+\cdots+Y_n=0.$$

The future m spins remain independent and each has expected value p.

The important point is that assuming the future spins have the same conditional win probability is not the error. Under the actual roulette model, they do have the same conditional win probability, namely p. The precise error occurred one step earlier, when the conditional expected average after observing L_n was incorrectly set equal to p for a finite horizon.

Therefore,

$$E[Y_{n+1}+\cdots+Y_{n+m}\mid L_n]=mp.$$

Hence,

$$E\left[\frac{Y_1+\cdots+Y_{n+m}}{n+m}\middle|L_n\right]=\frac{mp}{n+m}.$$

This is not equal to p for finite m.

Correct Conditional Expectation

The correct relation is

$$E[Y_{n+1}+\cdots+Y_{n+m}\mid L_n]=mp.$$

If

$$r=P(Y_{n+1}=1\mid L_n),$$

then independence gives

$$mr = mp.$$

Therefore,

$$r = p.$$

Hence,

$$P(Y_{n+1} = 1 \mid L_n) = \frac{18}{37}.$$

There is no increase in the probability of winning.

Misuse of the Law of Large Numbers

The incorrect argument also misinterprets the law of large numbers.

The law of large numbers says that

$$\frac{Y_1 + \cdots + Y_N}{N} \to p$$

as N becomes large.

It does not say that future probabilities must increase in order to compensate for past outcomes.

Suppose the first five spins are all losses.

If the next m spins behave normally, their expected number of wins is

$$mp.$$

The expected proportion over all $m + 5$ spins is

$$\frac{mp}{m+5}.$$

As

$$m \to \infty,$$

we have

$$\frac{mp}{m+5} = p\frac{m}{m+5} \to p.$$

Thus the overall proportion approaches p simply because the five initial losses become negligible relative to the growing number of later spins.

No short-run correction is required.

Impossible Probability Check

Another indication that the false formula cannot be correct is that

$$r = p\left(1 + \frac{n}{m}\right)$$

can produce probabilities larger than 1.

For example, if

$$n = 200$$

and

$$m = 100,$$

then

$$r = \frac{18}{37}\left(1 + \frac{200}{100}\right).$$

Therefore,

$$r = \frac{54}{37} \approx 1.459.$$

This would imply a probability of approximately

$$145.9\%,$$

which is impossible.

Thus the formula cannot represent a valid probability.

Check Against the Five Possible Errors

Conditional Probability Versus Intuition

The argument does not correctly compute

$$P(Y_{n+1} = 1 \mid L_n).$$

Instead, it replaces the conditional-probability calculation with the intuition that a deficit of wins must be corrected.

The correct calculation gives

$$P(Y_{n+1} = 1 \mid L_n) = p.$$

Long-Run Frequency Versus Short-Run Correction

The argument confuses long-run convergence with an immediate correction mechanism.

The law of large numbers describes averages over many trials. It does not imply that the next trial must compensate for previous losses.

Immediate Balancing

The argument assumes that the roulette wheel must quickly balance its past results.

Under the independent-spin model, no such mechanism exists.

Changing the Probability Model

The argument effectively changes the probability of the next spin from

$$p$$

to a larger value

$$r$$

without any physical or probabilistic change to the roulette wheel.

The next spin still has the same sample space and probability distribution.

Large Numbers Used to Make a False Next-Spin Claim

The statement

$$\frac{Y_1 + \cdots + Y_N}{N} \to p$$

is a valid long-run statement.

It does not imply

$$P(Y_{n+1} = 1 \mid L_n) > p.$$

The false proof incorrectly uses a true asymptotic statement to justify a false statement about one future spin.

Part IV: Throwing Good Money After Bad

For the loss-chasing experiment, I studied the classic Martingale strategy.

LLM Prompt

> Describe a Martingale betting strategy for European roulette in which the player bets on red and doubles the bet after each loss. Include a finite bankroll and a maximum table bet. Explain how the strategy operates and under what circumstances it can fail.

Martingale Strategy

The strategy is:

1. Start with a one-unit bet on red.
2. If the bet wins, end the session.
3. If the bet loses, double the next bet.
4. Continue doubling after each loss.
5. Stop if the required next bet cannot be funded by the bankroll.
6. Stop if the required bet exceeds the table maximum.

The simulation used

$$B = 100$$

as the initial bankroll,

$$b_0 = 1$$

as the base bet, and

$$M = 64$$

as the maximum table bet.

The possible sequence is

$$1, 2, 4, 8, 16, 32, 64, \ldots$$

With a bankroll of 100, the player can afford the sequence

$$1, 2, 4, 8, 16, 32.$$

The total loss after these six bets is

$$1 + 2 + 4 + 8 + 16 + 32 = 63.$$

The remaining bankroll is only

$$37,$$

so the next required bet of 64 cannot be placed.

Thus the session permits at most

$$k = 6$$

bets.

Why a Successful Session Earns One Unit

If the player loses the first j bets, the accumulated loss is

$$1 + 2 + \cdots + 2^{j-1} = 2^j - 1.$$

The next bet is

$$2^j.$$

If this bet wins, the net session profit is

$$2^j - (2^j - 1) = 1.$$

Therefore every successful Martingale session earns exactly

$$+1.$$

Probability of a Small Profit

Let

$$q = \frac{19}{37}$$

be the probability of losing one European roulette red bet.

The only way to fail is to lose all six permitted bets.

Therefore,

$$P(\text{large loss}) = q^6.$$

Thus,

$$P(\text{small profit}) = 1 - q^6.$$

Numerically,

$$P(\text{small profit}) = 1 - \left(\frac{19}{37}\right)^6 \approx 0.981664.$$

Therefore, the player wins one unit in approximately

$$98.17\%$$

of sessions.

Probability of a Large Loss

The failure probability is

$$P(\text{large loss}) = \left(\frac{19}{37}\right)^6 \approx 0.018336.$$

Thus the failure occurs only about

$$1.83\%$$

of the time.

However, the failure produces a loss of

$$63$$

units.

The session distribution is therefore approximately:

Outcome	Probability	Session Profit
Small profit	98.17%	$+1$
Large loss	1.83%	-63

Expected Profit of the Martingale Strategy

The expected profit is

$$E[\Pi] = (1 - q^6)(1) + q^6(-63).$$

Therefore,

$$E[\Pi] = 1 - 64q^6.$$

Substituting

$$q = \frac{19}{37}$$

gives

$$E[\Pi] = 1 - 64\left(\frac{19}{37}\right)^6.$$

Numerically,

$$E[\Pi] \approx -0.173522.$$

Thus the strategy loses money on average even though approximately 98.17% of sessions are profitable.

Monte Carlo Results

I simulated

$$1000000$$

independent Martingale sessions.

The results were:

Quantity	Theoretical	Simulated
Probability of small profit	0.981664	0.981576
Probability of large loss	0.018336	0.018424
Expected / mean profit	-0.173522	-0.179136

The theoretical variance of the session profit was approximately

$$73.7283.$$

The Monte Carlo standard error of the simulated mean was approximately

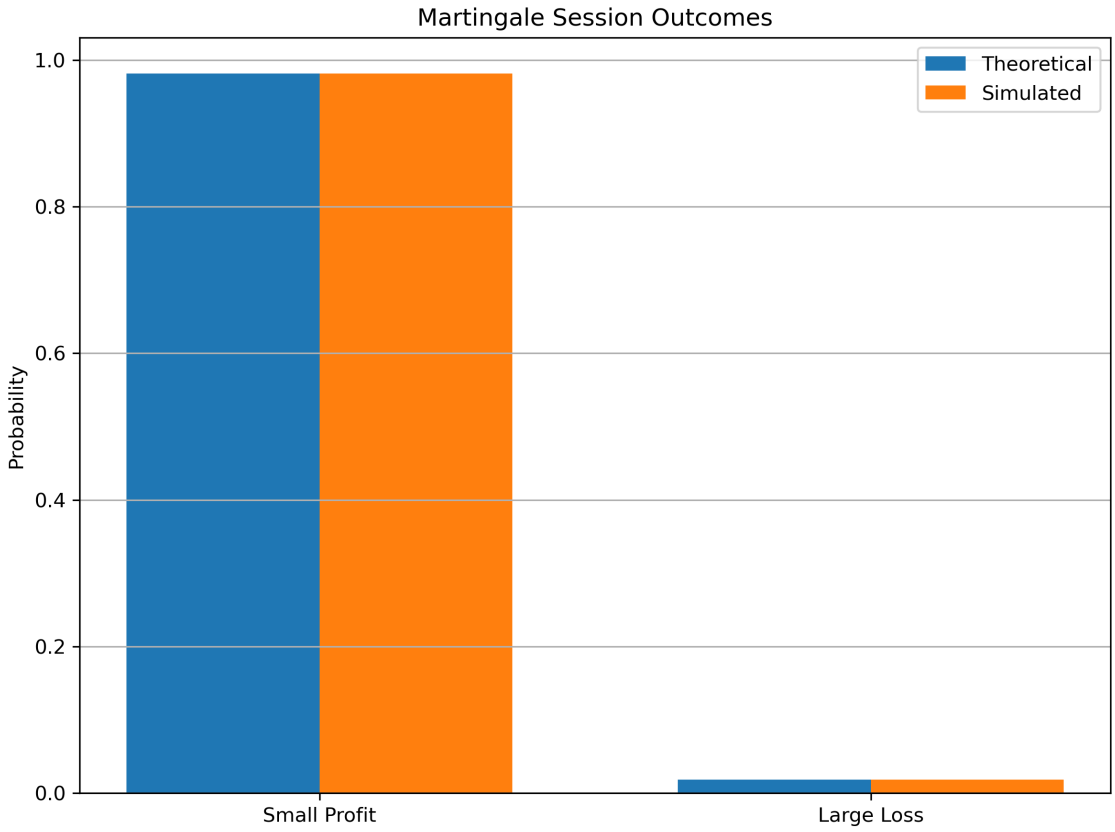
$$0.00859.$$

The difference between the theoretical and simulated mean profits was approximately

$$-0.654$$

standard errors.

Thus the simulation is consistent with the theoretical model.



Why Winning Often Can Still Produce Negative Expected Value

The expected contribution of successful sessions is approximately

$$(0.981664)(1) = 0.981664.$$

The expected contribution of failed sessions is approximately

$$(0.018336)(-63) \approx -1.15519.$$

Therefore,

$$0.981664 - 1.15519 \approx -0.173522.$$

The rare large losses outweigh the many small wins.

Thus,

$$P(\Pi > 0) \approx 0.9817$$

while

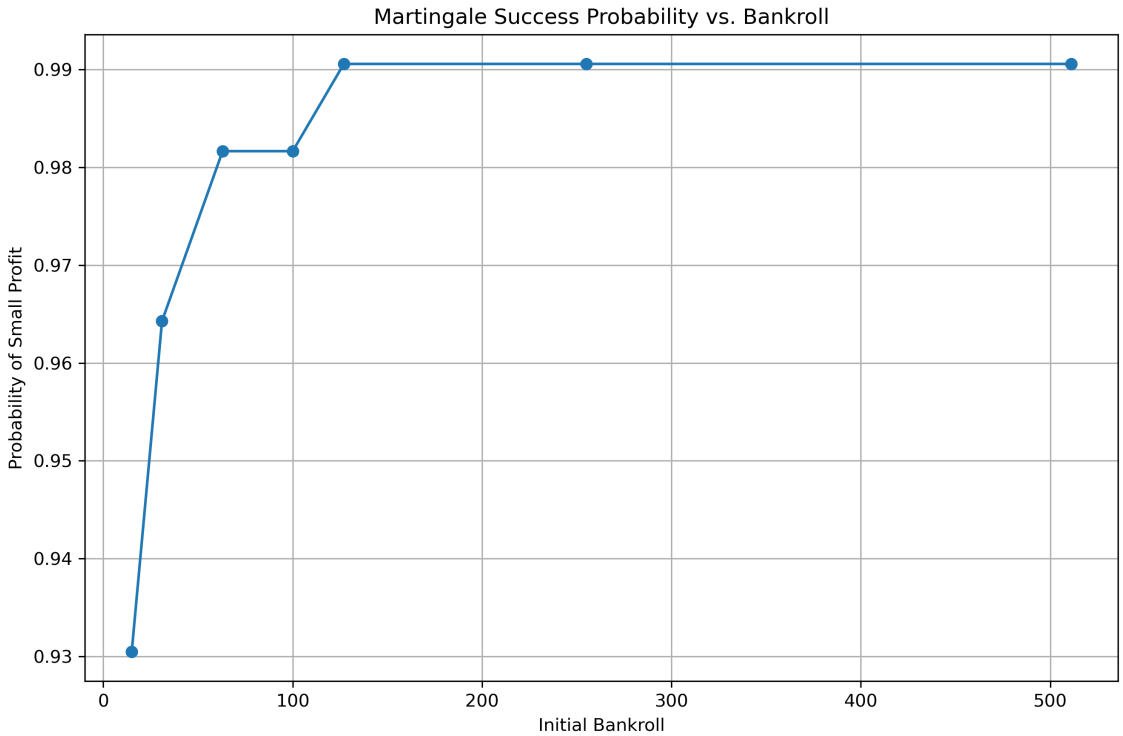
$$E[\Pi] < 0.$$

This demonstrates that a high probability of winning does not imply positive expected value.

Effect of a Finite Bankroll

The simulation also tested several different bankroll sizes.

Bankroll	Allowed Bets	Small-Profit Probability	Large-Loss Probability	Large Loss	Expected Profit
15	4	0.93046	0.06954	-15	-0.11257
31	5	0.96429	0.03571	-31	-0.14264
63	6	0.98166	0.01834	-63	-0.17352
100	6	0.98166	0.01834	-63	-0.17352
127	7	0.99058	0.00942	-127	-0.20524
255	7	0.99058	0.00942	-127	-0.20524
511	7	0.99058	0.00942	-127	-0.20524



Increasing the bankroll increases the probability of obtaining the small profit because the player can survive a longer sequence of losses.

However, it also makes the possible failure larger.

For example, four allowed bets risk

$$15$$

units, while seven allowed bets risk

$$127$$

units.

Thus increasing the bankroll does not remove the negative expected value.

Effect of the Table Maximum

The table maximum in the experiment was

$$64.$$

Once the bankroll is large enough to fund

$$1, 2, 4, 8, 16, 32, 64,$$

the next required Martingale bet would be

$$128.$$

This exceeds the table maximum.

Therefore, additional bankroll cannot create another doubling step.

More generally, with base bet 1, bankroll B, and maximum table bet M, the number of complete Martingale bets that can actually be placed is

$$k = \min\left(\lfloor \log_2(B+1) \rfloor, 1 + \lfloor \log_2 M \rfloor\right).$$

The first term comes from the bankroll requirement

$$1 + 2 + \cdots + 2^{k-1} = 2^k - 1 \le B,$$

and the second comes from the table-limit requirement

$$2^{k-1} \le M.$$

For $M = 64$, at most seven Martingale bets can be placed, regardless of how large the bankroll becomes.

This explains why bankrolls

$$127,\ 255,\ \text{and } 511$$

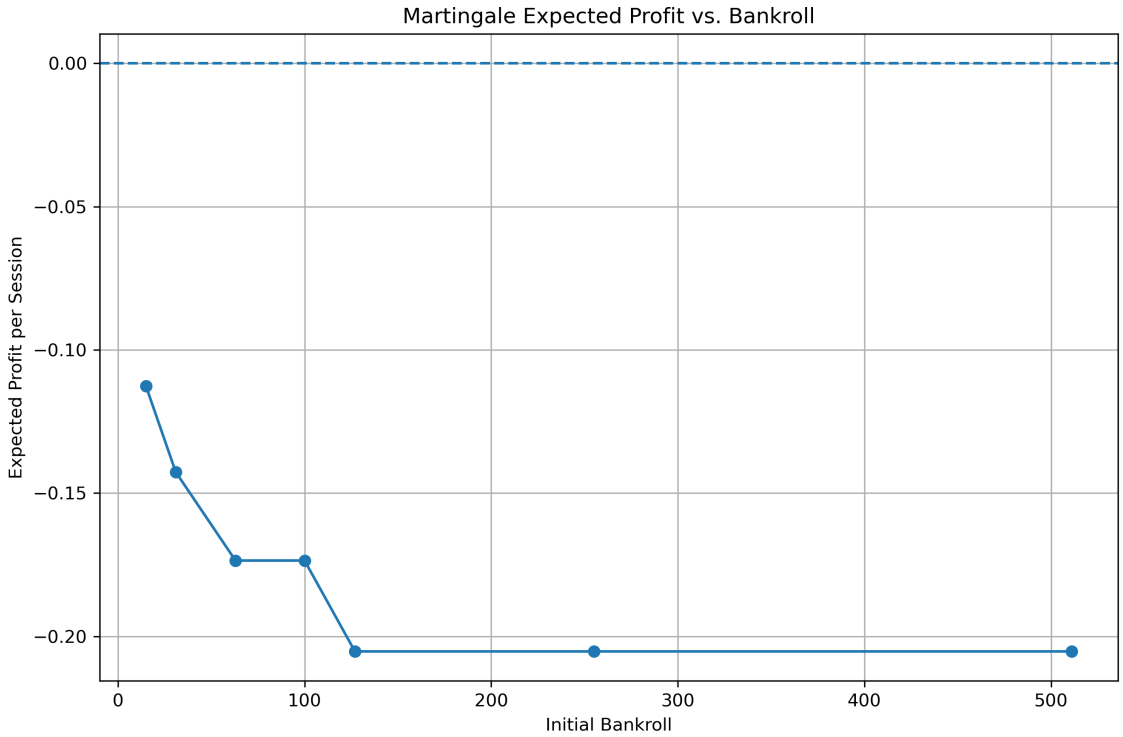
produce identical results.

Their probability of a small profit remains approximately

$$0.990584$$

and their expected profit remains approximately

$$-0.205239.$$



General Martingale Expected Value

If the player can make k complete doubling bets, then

$$P(\text{failure}) = q^k.$$

The loss after k consecutive losses is

$$2^k - 1.$$

Therefore,

$$E[\Pi] = (1 - q^k) - (2^k - 1)q^k.$$

Simplifying,

$$E[\Pi] = 1 - 2^k q^k.$$

Hence,

$$E[\Pi] = 1 - (2q)^k.$$

For European roulette,

$$2q = 2\left(\frac{19}{37}\right) = \frac{38}{37} > 1.$$

Therefore,

$$E[\Pi] < 0.$$

Thus the Martingale strategy does not remove the casino advantage.

There is also a useful expectation check that does not depend on the closed-form Martingale formula. Whenever a bet of size b is actually placed on red, the next spin is independent of the past and its conditional expected profit is

$$-\frac{b}{37}.$$

Therefore every additional wager made by the strategy has negative conditional expectation. Summing over the wagers that are actually placed cannot turn the house edge into a positive expected return. The Martingale changes the distribution of outcomes, but not the sign of the expected value.

It changes the distribution of outcomes into many small wins and rare large losses.

Part V: Reflection

Was the LLM Better as a Tutor or as a Generator of Plausible Mistakes?

The LLM was more useful as a tutor, but this homework also demonstrated how effectively it can generate convincing mathematical mistakes.

As a tutor, the LLM helped define the roulette probability spaces, construct random variables, derive expected values, formulate conditional probabilities, and write simulations.

For example, it correctly identified

$$P_E(\text{win}) = \frac{18}{37}$$

and

$$E[X^E] = -\frac{1}{37}.$$

It was also useful in constructing the Martingale simulation and comparing theoretical results with Monte Carlo outcomes.

However, Part III showed that mathematically sophisticated language is not enough to guarantee a correct argument.

When asked to defend the gambler's fallacy, the LLM produced an argument containing valid ideas such as expectation and the law of large numbers but combined them incorrectly.

Therefore, I found the LLM most useful as a tutor when its output could be independently checked.

What Was the Most Convincing Wrong Argument?

The most convincing false argument was the claim that a losing streak creates a deficit of wins that future spins must correct.

The LLM began with the valid statement that the long-run proportion of wins should approach

$$p = \frac{18}{37}.$$

It then incorrectly assumed

$$E\left[\frac{Y_1 + \cdots + Y_{n+m}}{n+m} \backslash \text{midL}_n\right] = p.$$

From this, it derived

$$P(Y_{n+1} = 1 \mid L_n) = p\left(1 + \frac{n}{m}\right).$$

For five losses and $m = 100$, it obtained the false prediction

$$P(\text{win next} \mid 5 \text{ losses}) \approx 51.08\%.$$

The argument was convincing because most of its mathematical notation and terminology were valid. The error was hidden inside the conditional-expectation assumption.

How Did Conditional Probability Expose the Error?

Conditional probability gives

$$P(Y_{n+1} = 1 \mid L_n) = \frac{P(Y_{n+1} = 1 \cap L_n)}{P(L_n)}.$$

By independence,

$$P(Y_{n+1} = 1 \cap L_n) = P(Y_{n+1} = 1)P(L_n).$$

Therefore,

$$P(Y_{n+1} = 1 \mid L_n) = P(Y_{n+1} = 1).$$

Hence,

$$P(Y_{n+1} = 1 \mid L_n) = \frac{18}{37}.$$

The previous losing streak has no effect on the probability of the next spin.

The law of large numbers does not require future spins to compensate for previous deviations. The influence of a fixed early losing streak simply becomes negligible as more independent spins occur.

What Does This Teach About LLM Reliability?

This homework showed that an LLM can be a useful mathematical tool, but its output should be treated as a proposed argument rather than as automatically correct mathematics.

The LLM was useful for:

- organizing probability models;
- defining random variables;
- deriving equations;
- suggesting simulations;
- explaining mathematical concepts;
- producing alternative arguments.

However, it was also able to construct a false proof using correct mathematical terminology.

The most important lesson is that definitions, assumptions, conditioning, and logical implications must still be checked independently.

The Monte Carlo experiments were useful as an additional check, but numerical simulation also does not replace mathematical reasoning.

A useful workflow is

$$\text{LLM suggestion} \rightarrow \text{identify assumptions} \rightarrow \text{derive the mathematics} \rightarrow \text{check critical ste}$$

Thus the LLM is useful for accelerating mathematical work, but it should not be treated as an unquestionable source of mathematical truth.

Conclusion

This homework demonstrates two related mistakes in gambling intuition.

First, the gambler's fallacy incorrectly assumes that previous losses increase the probability of a future win.

For European roulette,

$$P(X_{n+1} > 0 \mid L_n) = P(X_{n+1} > 0) = \frac{18}{37}.$$

The Monte Carlo experiments confirmed this for losing streaks of several different lengths.

Second, the Martingale strategy attempts to respond to losses by increasing the amount wagered.

For the baseline experiment,

$$P(\text{small profit}) \approx 98.17\%,$$

but

$$E[\Pi] \approx -0.17352.$$

Thus a strategy can win very frequently while still having negative expected value.

The LLM was useful for building the models and simulations, but Part III demonstrated that it can also produce a mathematically plausible false argument.

The main lesson is that probabilistic reasoning should be based on explicit probability models, conditional probabilities, and expected values rather than intuition or persuasive mathematical language.